

Portfolio

Madhulata Solutions

Artificial Intelligence | Software Development | Automation | Digital Transformation

Company Portfolio

Madhulata Solutions develops Artificial Intelligence applications, business automation systems, modern web applications, and enterprise software solutions.

Our portfolio demonstrates expertise in solving business challenges through scalable software engineering and AI-powered technologies.

Project 1

AI Company Knowledge Assistant

Category

Artificial Intelligence

Description

An AI-powered assistant that answers company-related questions by retrieving information from internal documents using Retrieval-Augmented Generation (RAG).

Features

- **Document Search**
- **Semantic Search**
- **AI-generated Answers**
- **PDF Support**
- **Word Document Support**
- **Knowledge Base**
- **Context-aware Responses**

Technologies

- **Python**
- **FastAPI**
- **LangChain**
- **ChromaDB**
- **OpenAI Compatible LLM**
- **HTML**

- **CSS**
- **JavaScript**

Business Benefits

- **Reduces employee search time**
 - **Improves customer support**
 - **Centralizes company knowledge**
 - **Enables intelligent document search**
-

Project 2

AI Customer Support Chatbot

Category

Artificial Intelligence

Description

A conversational AI chatbot integrated with business websites to answer customer queries and assist visitors.

Features

- **Natural Language Conversations**
- **FAQ Automation**
- **Lead Collection**
- **Product Recommendations**
- **Website Integration**
- **Human Handoff Support**

Technologies

- **Python**
- **FastAPI**
- **LangChain**
- **OpenAI API**
- **HTML**
- **CSS**
- **JavaScript**

Business Benefits

- **24×7 Customer Support**
 - **Improved Customer Experience**
 - **Reduced Support Costs**
 - **Faster Response Time**
-

Project 3

Workflow Automation Platform

Category

Business Automation

Description

An automation platform that connects multiple business applications and automates repetitive workflows.

Features

- **Email Automation**
- **Lead Processing**
- **Calendar Scheduling**
- **CRM Integration**
- **Notification System**
- **Approval Workflows**

Technologies

- **n8n**
- **REST APIs**
- **Webhooks**
- **Gmail API**
- **Google Calendar API**

Business Benefits

- **Saves manual effort**
 - **Faster business operations**
 - **Improved accuracy**
 - **Better team collaboration**
-

Project 4

Lead Management System

Category

Web Application

Description

A centralized platform for managing customer inquiries and tracking leads throughout the sales process.

Features

- **Contact Forms**
- **Lead Tracking**
- **Status Management**
- **Customer Database**
- **Admin Dashboard**
- **Analytics**

Technologies

- **FastAPI**
- **PostgreSQL**
- **HTML**
- **CSS**
- **JavaScript**

Business Benefits

- **Organized customer data**
- **Better follow-up management**
- **Improved sales tracking**

Project 5

Company Website with AI Assistant

Category

Web Development

Description

A modern business website integrated with an AI assistant capable of answering customer questions and collecting project requirements.

Features

- **Responsive Design**
- **AI Chatbot**
- **Contact Forms**
- **Service Pages**
- **Portfolio**
- **Blog Section**
- **SEO Optimization**

Technologies

- **HTML**
- **CSS**
- **JavaScript**
- **FastAPI**
- **LangChain**

Business Benefits

- **Better online presence**
- **Higher customer engagement**
- **Automated lead generation**

Project 6

Business Analytics Dashboard

Category

Data Analytics

Description

An interactive dashboard that provides business insights using charts, reports, and KPIs.

Features

- **Sales Dashboard**
- **KPI Monitoring**
- **Interactive Reports**
- **Export Reports**
- **Business Analytics**

Technologies

- Python
- SQL
- PostgreSQL
- JavaScript

Business Benefits

- Better decision making
 - Improved reporting
 - Data-driven business strategy
-

Project 7

Document Management System

Category

Enterprise Software

Description

A secure document management platform for storing, searching, and managing organizational documents.

Features

- Document Upload
- Version Control
- Search
- User Permissions
- Document Categories

Technologies

- Python
- FastAPI
- PostgreSQL

Business Benefits

- Secure document storage
- Faster document retrieval
- Improved collaboration

Project 8

AI Resume Screening System

Category

Artificial Intelligence

Description

An AI-powered recruitment assistant that analyzes resumes and ranks candidates based on job descriptions.

Features

- **Resume Parsing**
- **Candidate Ranking**
- **Skill Matching**
- **Automated Shortlisting**

Technologies

- **Python**
- **LangChain**
- **FastAPI**

Business Benefits

- **Faster recruitment**
 - **Better candidate matching**
 - **Reduced manual effort**
-

Project 9

Inventory Management System

Category

Business Software

Description

A web-based inventory management application designed to monitor stock levels, suppliers, and purchase orders.

Features

- **Product Management**
- **Stock Tracking**

- **Supplier Records**
- **Reports**
- **Alerts**

Technologies

- **Python**
- **PostgreSQL**
- **HTML**
- **CSS**
- **JavaScript**

Business Benefits

- **Inventory optimization**
- **Reduced stock shortages**
- **Better purchasing decisions**

Technical Expertise Demonstrated

The projects in our portfolio demonstrate expertise in:

- **Artificial Intelligence**
- **Generative AI**
- **Retrieval-Augmented Generation (RAG)**
- **AI Chatbots**
- **LangChain**
- **Workflow Automation**
- **FastAPI Development**
- **REST API Development**
- **PostgreSQL**
- **ChromaDB**
- **HTML**
- **CSS**
- **JavaScript**
- **Responsive Web Applications**
- **Database Design**

- **Cloud-Ready Applications**
-

Development Approach

Each project follows a structured development process:

- 1. Requirement Analysis**
 - 2. Solution Design**
 - 3. Architecture Planning**
 - 4. Development**
 - 5. Testing**
 - 6. Client Review**
 - 7. Deployment**
 - 8. Maintenance**
-

Industries Served

The solutions are designed for businesses in:

- **Manufacturing**
 - **Healthcare**
 - **Education**
 - **Retail**
 - **Finance**
 - **Logistics**
 - **Hospitality**
 - **Real Estate**
 - **Information Technology**
 - **Professional Services**
-

Frequently Asked Questions

Does Madhulata Solutions develop AI applications?

Yes. The company develops AI chatbots, RAG systems, AI assistants, and intelligent automation solutions.

Can the company build custom software?

Yes. All software solutions are customized according to business requirements.

Which technologies are used?

Python, FastAPI, LangChain, PostgreSQL, ChromaDB, HTML, CSS, JavaScript, Docker, and n8n.

Does Madhulata Solutions provide workflow automation?

Yes. Workflow automation is implemented using n8n, APIs, and webhooks.

Can existing systems be integrated?

Yes. Existing ERP, CRM, websites, and third-party services can be integrated using secure APIs.

Note

Some portfolio items represent sample, prototype, or demonstration projects created to showcase Madhulata Solutions' technical capabilities and approach to solving business problems. Actual implementations are customized according to each client's requirements.

Tip for Your RAG Knowledge Base

Since you're building an AI assistant for Madhulata Solutions, I recommend organizing your knowledge base into 10 PDFs:

- 1. About Madhulata Solutions**
- 2. Company Profile**
- 3. Services**
- 4. Portfolio**
- 5. Technology Stack**
- 6. Pricing & Engagement Models**
- 7. Frequently Asked Questions (FAQ)**
- 8. Company Policies**
- 9. Contact Information**
- 10. Client Onboarding Guide**

This structure makes retrieval more accurate and gives your LangChain-based RAG system comprehensive coverage of your business.